

```
[15]: import pandas as pd
a=pd.Series([1,2,3.6,4,6])
print(a) #it prints both value and index
print(a.values) #it prints only values
print(a.index) #it prints like index value
print(a.dtype) #data type alone

import pandas as pd
a=pd.Series([True,False,False])
print(a) #it prints both value and index
print(a.values) #it prints only values
print(a.index) #it prints like index value
print(a.dtype) #data type alone

import pandas as pd
a=pd.Series([True,6,6.8])
print(a) #it prints both value and index
print(a.values) #it prints only values
print(a.index) #it prints like index value
print(a.dtype) #data type alone

import pandas as pd
a=pd.Series(['priya','rithu','raju','rithan','suma'])
print(a) #it prints both value and index
print(a.values) #it prints only values
print(a.index) #it prints like index value
print(a.dtype) #data type alone

#assessing elements
print(a[1])
#assessing elements using location
print(a.loc[1]) #Location
print(a.iloc[1]) #index location
#it is
```

```
#slicing
print(a[1:3])
print(a.loc[1:3]) #this will take last index value also
print(a.iloc[2:5]) # in this the last value will be -1
```

```
0    1.0
1    2.0
2    3.6
3    4.0
4    6.0
dtype: float64
[1.  2.  3.6 4.  6. ]
RangeIndex(start=0, stop=5, step=1)
float64
0     True
1    False
2    False
dtype: bool
[ True False False]
RangeIndex(start=0, stop=3, step=1)
bool
0     True
1         6
2     6.8
dtype: object
[True 6 6.8]
RangeIndex(start=0, stop=3, step=1)
object
0    priya
1    rithu
2     raju
3   rithan
4     suma
dtype: object
['priya' 'rithu' 'raju' 'rithan' 'suma']
RangeIndex(start=0, stop=5, step=1)
```

```

RangeIndex(start=0, stop=5, step=1)
object
rithu
rithu
rithu
1    rithu
2    raju
dtype: object
1    rithu
2    raju
3    rithan
dtype: object
2    raju
3    rithan
4    suma
dtype: object

```

```

[57]: import pandas as pd
arr=pd.Series([1,2,3,4,56,7],index=['A','B','C','D','E','F']) #we are creating a respective index
print(arr[['A','C','F']])
print(arr)
print(arr['E']) #by acessing this
#vectorized operation *(no loop required)
print(arr+5)
print(arr.iloc[1]) #accessed by position like 0 1 2
print(arr.loc['A']) #it accessed by Labelled value like A B C

#Stastical function
print(arr.sum())
print(arr.std())
print(arr.max())
print(arr.min())
print(arr.mean())
print(arr.median())

```



```

#Boolean indexing
print(arr[arr>30])

print(len(arr)) #Lenth of array
print(arr.value_counts()) #to count individual values

#string operation
name=pd.Series(['jack','vishnu','anitha','sashmitha','srinivas'])
print(name.str.upper()) #to convert the string into uppercase
print(name.str.len()) #to identify the length of each string

```

```

A      1
C      3
F      7
dtype: int64
A      1
B      2
C      3
D      4
E     56
F      7
dtype: int64
56
A      6
B      7
C      8
D      9
E     61
F     12
dtype: int64
2
1
73
21.57235885726609
56
1

```

```
1
73
21.57235885726609
56
1
12.166666666666666
3.5
```

```
E      56
dtype: int64
6
```

```
1      1
2      1
3      1
4      1
56     1
7      1
```

```
Name: count, dtype: int64
```

```
0      JACK
1      VISHNU
2      ANITHA
3      SASHMITHA
4      SRINIVAS
```

```
dtype: object
```

```
0      4
1      6
2      6
3      9
4      8
```

```
dtype: int64
```

```
[46]: import pandas as pd
a=pd.Series({
    "gayathri":16,
    "rithanyaa":38,
    "bavanthika":8})
print(a)
print(a["rithanyaa"])
```

```
gayathri      16
rithanyaa     38
bavanthika     8
dtype: int64
38
```

```
[58]: s1=pd.Series([10,20,30],index=['A','B','C'])
s2=pd.Series([1,2,3],index=['B','C','D'])
print(s1+s2)
s=pd.Series([1,2,3,5,5])
print(s.count())
print(s.unique()) #it shows only unique values not repeated values if repeated it print only once
print(s.nunique()) #it shows the unique value count
```

```
A      NaN
B      21.0
C      32.0
D      NaN
dtype: float64
5
[1 2 3 5]
4
```

```
[181]: import pandas as pd
#method1 : from dictionary
data={
    'name':['monica','nagul','prem'],
    'rollno':[27,20,24],
    'Age':[19,20,21]}
df=pd.DataFrame(data)
print(df)

#from series to dataframe
s1=pd.Series([1,2,3,4,5])
s2=pd.Series([5.6,9.54,9.4,54.6,6])
s3=pd.Series([True,False,True,False])
df=pd.DataFrame({'sno':s1,
                 'cgpa':s2,
                 'attendance':s3})

print(df)

#list of list
data=[['athish',22,98],
      ['rithikaa',20,89],
      ['prathiksha',20,97]]
df=pd.DataFrame(data,columns=["name","age","mark1"])
print(df)

#list of dictionary
data=[
    {"name":"sakthi","course":"data science","grade":'A'},
    {"name":"madhu","course":"Aiml","grade":'A+'},
    {"name":"ragavi","course":"Aiml","grade":'A++'}]
df=pd.DataFrame(data,index=["std1","std2","std3"])
print(df)
```

```
#adding one extra column
df['marks']=[98,97,96]
print(df)

#adding one extra row
std4=pd.DataFrame([{"name":"rithu","course":"Aiml","grade":"B","marks":90}],index=["std4"])
df=pd.concat([df,std4])
print(df)

#create multiple rows
students=pd.DataFrame([{"name":"sanjeev","course":"csda","grade":"A","marks":98},
                        {"name":"divya","course":"aiml","grade":"0+","marks":99}],index=["std5","std6"])
df=pd.concat([df,students])
print(df)

#particular column
print(df.course)
print(df[["grade","marks"]])

#shape attribute
print(df.shape)
#find no of rows
print(df.shape[0])
#find no of columns
print(df.shape[1])

#Dtype attribute
#used to check data type of all columns
print(df.dtypes)

#info
print(df.info())
print(df.describe())
```



```

#select specofic row
print(df.loc['std1'])
print(df.loc[['std1','std4']])
print(df.iloc[3])

#update value
df.loc['std3','grade']='B++'
print(df)

#delete column
df=df.drop('grade',axis=1)
print(df)

#delete but store
df.drop('std3',axis=0,inplace=True)
print(df)

#sorting
df.sort_values('marks',inplace=True)
print(df)
df.sort_values('marks',ascending=False,inplace=True)
print(df)

#stastical function for particular column
print(df['marks'].max())

```

	name	rollno	Age
0	monica	27	19
1	nagul	20	20
2	prem	24	21

	sno	cgpa	attandance
0	1	5.60	True
1	2	9.54	False
2	3	9.40	True
3	4	54.60	False

4	5	6.00	NaN
	name	age	mark1
0	athish	22	98
1	rithikaa	20	89
2	prathiksha	20	97

	name	course	grade
std1	sakthi	data science	A
std2	madhu	Aiml	A+
std3	ragavi	Aiml	A++

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	Aiml	A+	97
std3	ragavi	Aiml	A++	96

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	Aiml	A+	97
std3	ragavi	Aiml	A++	96
std4	rithu	Aiml	B	90

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	Aiml	A+	97
std3	ragavi	Aiml	A++	96
std4	rithu	Aiml	B	90
std5	sanjeev	csda	A	98
std6	divya	aiml	0+	99

std1	data science
std2	Aiml
std3	Aiml
std4	Aiml
std5	csda
std6	aiml

Name: course, dtype: object

	grade	marks
std1	A	98
std2	A+	97
std3	A++	96

```

std4      B      90
std5      A      98
std6      0+     99
(6, 4)
6
4
name      object
course    object
grade     object
marks     int64
dtype: object
<class 'pandas.core.frame.DataFrame'>
Index: 6 entries, std1 to std6
Data columns (total 4 columns):
#   Column  Non-Null Count  Dtype
---  -
0   name    6 non-null      object
1   course  6 non-null      object
2   grade   6 non-null      object
3   marks   6 non-null      int64
dtypes: int64(1), object(3)
memory usage: 240.0+ bytes
None

```

```

              marks
count    6.000000
mean    96.333333
std      3.265986
min     90.000000
25%     96.250000
50%     97.500000
75%     98.000000
max     99.000000
name                    sakthi
course      data science
grade              A
marks              98

```


4 Joanna Slovakia (Slovak Republic)

First Name Country

95 Karl Guyana

96 Lynn Sri Lanka

97 Fred Solomon Islands

98 Yvonne Aruba

99 Clarence Honduras

First Name

Country

0 Sheryl Chile

1 Preston Djibouti

2 Roy Antigua and Barbuda

3 Linda Dominican Republic

4 Joanna Slovakia (Slovak Republic)

5 Aimee Bosnia and Herzegovina

6 Darren Pitcairn Islands

7 Brett Bulgaria

8 Sheryl Cyprus

9 Michelle Timor-Leste

10 Carl Guernsey

11 Jenna Vietnam

12 Tracey Togo

13 Kristine Sri Lanka

14 Faith Singapore

15 Miranda Oman

16 Caroline Western Sahara

17 Greg Mozambique

18 Clifford South Georgia and the South Sandwich Islands

19 Joanna French Polynesia

20 Maxwell Malta

21 Kiara Netherlands

22 Colleen Paraguay

23 Janet Lao People's Democratic Republic

24 Shane Albania

25 Marcus Panama

26 Dakota Belarus

27 Frederick Switzerland

28	Stefanie	Saint Vincent and the Grenadines
29	Kent	Tanzania
30	Jack	Zimbabwe
31	Tom	Denmark
32	Gabriel	Liechtenstein
33	Kaitlyn	United States of America
34	Faith	Bahamas
35	Tammie	Belize
36	Nicholas	Uruguay
37	Jordan	Solomon Islands
38	Bruce	Montenegro
39	Sherry	Poland
40	Natalie	Dominican Republic
41	Bryan	Burkina Faso
42	Wayne	Bolivia
43	Luis	Bulgaria
44	Rhonda	Monaco
45	Joanne	Palau
46	Geoffrey	Uzbekistan
47	Gloria	Ghana
48	Brady	United Arab Emirates
49	Latoya	Belarus
50	Gerald	Canada
51	Samuel	Algeria
52	Patricia	Swaziland
53	Stacie	Madagascar
54	Robin	Ecuador
55	Ralph	Palestinian Territory
56	Phyllis	Saint Barthelemy
57	Danny	United Arab Emirates
58	Kathy	South Georgia and the South Sandwich Islands
59	Kelli	Sao Tome and Principe
60	Lynn	Portugal
61	Shelley	Togo
62	Eddie	Ethiopia
63	Chloe	Netherlands

64	Eileen	Liberia
65	Fernando	Lithuania
66	Makayla	New Caledonia
67	Tom	Kiribati
68	Virginia	French Southern Territories
69	Riley	Canada
70	Alexandria	Iran
71	Richard	Morocco
72	Candice	Zimbabwe
73	Anita	Russian Federation
74	Regina	Solomon Islands
75	Debra	United States Virgin Islands
76	Brittany	Kyrgyz Republic
77	Cassidy	Myanmar
78	Laurie	Dominica
79	Alejandro	Iceland
80	Leslie	Micronesia
81	Kathleen	Saint Vincent and the Grenadines
82	Hunter	Isle of Man
83	Chad	Oman
84	Corey	Fiji
85	Emma	New Zealand
86	Duane	Maldives
87	Alison	Benin
88	Vernon	Yemen
89	Lori	Namibia
90	Nina	Bhutan
91	Shane	Hungary
92	Collin	Anguilla
93	Sherry	Solomon Islands
94	Darrell	Mali
95	Karl	Guyana
96	Lynn	Sri Lanka
97	Fred	Solomon Islands
98	Yvonne	Aruba
99	Clarence	Honduras

```
99      Clarence                                     Honduras
0              Chile
1              Djibouti
2      Antigua and Barbuda
3      Dominican Republic
4      Slovakia (Slovak Republic)
      ...
95              Guyana
96              Sri Lanka
97      Solomon Islands
98              Aruba
99              Honduras
Name: Country, Length: 100, dtype: object
```
